

Supplemental material for “Time-domain characterization and correction of on-chip distortion of control pulses in a quantum processor”

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This supplement provides experimental details and derivations supporting claims made in the main text. First, we describe the experimental setup. We then discuss the limitations of the Cryoscope, showing how undesired distortions are suppressed for a typical step response and how the nonlinear response of the qubit to flux helps in reconstructing the step response. The third section details a simple model that describes the signal-to-noise ratio of the experiment. Next, we provide details on the hardware implementations of the FIR and IIR filters used to correct distortions in real time. Finally, we provide experimental data demonstrating the ability to use Cryoscope to reconstruct an arbitrary signal.

I. DEVICE AND EXPERIMENTAL SETUP

The data shown in this letter were acquired using two devices mounted in different dilution refrigerators. In all experiments, a Zurich Instruments HDAWG equipped with real-time digital filters was used to generate the flux pulses. The output of the AWG was connected to the RF port of a Mini-Circuits ZFBT 6GW+ bias tee while the DC port was connected to a DC current source. The RF+DC port of the bias tee was connected to the flux control line entering the fridge. The flux control line contains a 20 dB attenuator at the 4 K stage as well as a Mini-Circuits VLFX1050 low-pass filter and a homebuilt eccosorb filter before being connected to the flux control line on the device. The control-line coaxial cables between 4 K and mixing chamber plate were superconducting (NbTi, inner and outer conductor) for one device and stainless steel (inner and outer) for the other. In all cases it was possible to correct distortions to within $\sim 0.1\%$.

II. LIMITATIONS OF THE CRYOSCOPE

In this section we first investigate the accuracy of the Cryoscope for a physically motivated step response including distortions due to AWG bandwidth, bias tee, skin effect, and on-chip response. We show that the inaccuracy is small using typical distortion parameters. Next, we investigate the effects of a single-pole low-pass filter for which the error in the reconstruction is significant on the timescale of the filter.

Our analysis is based on a numerical calculation of the acquired relative phase φ_τ , yielding a noiseless Cryoscope measurement. Specifically,

$$\varphi_\tau = 2\pi \int_0^\infty a \left[\Phi_s(s(t) - s(t - \tau)) \right]^k dt, \quad (S1)$$

where Φ_s is the amplitude of the applied square flux pulse, $s(t)$ is the step response and a parametrizes the dependence between magnetic flux Φ and qubit detuning in the quadratic approximation: $\Delta f_Q = a\Phi^k$ with $k = 2$. Because the phases are calculated in simulation it is possible to extract the contributions to $\overline{\Delta f_R}$:

$$\overline{\Delta f_R}(\tau, \Delta\tau) \equiv \frac{\varphi_{\tau+\Delta\tau} - \varphi_\tau}{2\pi\Delta\tau} = \underbrace{\frac{1}{\Delta\tau} \int_\tau^{\tau+\Delta\tau} \Delta f_Q(\Phi_{Q,\tau+\Delta\tau}(t)) dt}_{\overline{\Delta f_Q}} + \underbrace{\frac{1}{\Delta\tau} \int_{\tau+\Delta\tau}^{\tau_{\text{sep}}} \Delta f_Q(\Phi_{Q,\tau+\Delta\tau}(t)) dt}_{\varepsilon_{\tau+\Delta\tau, \text{Off}}} - \underbrace{\frac{1}{\Delta\tau} \int_\tau^{\tau_{\text{sep}}} \Delta f_Q(\Phi_{Q,\tau}(t)) dt}_{\varepsilon_{\tau, \text{Off}}}. \quad (S2)$$

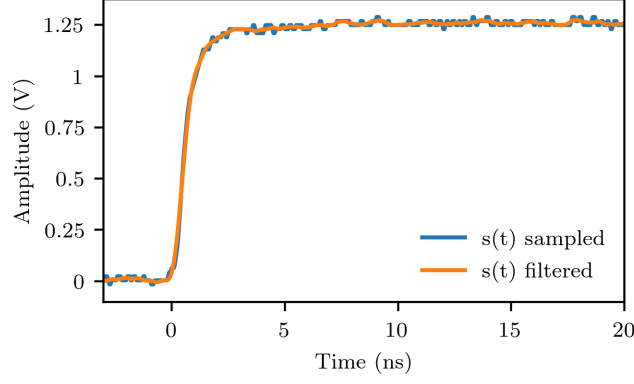


FIG. S1. Measured step response of the HDAWG in amplified mode, measured using a Rohde & Schwarz RTO1024 oscilloscope (blue). The signal is filtered with a Savitsky-Golay filter (orange) in order to determine the impulse response from the derivative.

A. Cryoscope reconstruction of a typical step response

An overview of the distortion models used can be found in Table S1. The response of the HDAWG is taken into account by performing a convolution with an impulse response extracted from the measured step response. This step response was measured when the HDAWG was operated in amplified mode and shown in Fig. S1. The step response of the bias tee is modeled as a single exponential high-pass filter of the form $s(t) = e^{-t/\tau_{\text{HP}}} \cdot u(t)$ and several exponential filters of the form $s(t) = 1 + Ae^{-t/\tau_{\text{IR}}} \cdot u(t)$, where τ_{IR} and τ_{HP} are the relevant time constants, A is an amplitude coefficient and $u(t)$ is the Heaviside step function. The coefficients used are based on a measured step response for a bias tee and are the same as in Ref. S1. We note that the coefficients are known to vary slightly between different bias tees of the same model. The skin effect is modeled according to Ref. S2 with an attenuation of $\alpha_{\text{GHz}} = 2.1$ dB at 1 GHz. The signal is filtered with a Savitsky-Golay filter in order to determine the derivative and the impulse response.

Effect	Model	Model parameters	Notes
AWG response	$\star h_{\text{AWG}}$	-	Measured step response of HDAWG in amplified mode (Fig. S1)
Bias tee	$s(t) = (e^{-t/\tau_{\text{HP}}}) \cdot u(t)$	$\tau_{\text{HP}} = 41 \mu\text{s}$	Mini-Circuits ZFBT-6GW+
Bias tee	$s(t) = (1 + A \cdot e^{-t/\tau_{\text{IR}}}) \cdot u(t)$	$\tau_{\text{IR}} = 15 \mu\text{s}$ $A = 0.13$	Ref. S1
Bias tee	$s(t) = (1 + A \cdot e^{-t/\tau_{\text{IR}}}) \cdot u(t)$	$\tau_{\text{IR}} = 6.4 \mu\text{s}$ $A = 0.99$	Ref. S1
Skin effect	$s(t) = (1 - \text{erfc}(\alpha_{\text{GHz}}/21\sqrt{t})) \cdot u(t)$	$\alpha_{\text{GHz}} = 2.1\text{dB}$	Model according to Ref. S2
On-chip response	$s(t) = (1 + A \cdot e^{-t/\tau_{\text{IR}}}) \cdot u(t)$	$\tau_{\text{IR}} = 2 \text{ ns}$ $A = 0.6$	

TABLE S1. Overview of the distortion models used and their coefficients. Fig. S2 illustrates the cumulative influence of all listed effects and their reconstruction using Cryoscope.

We model the effect of distortions on square pulses (truncated at time τ and $\tau + \Delta\tau$) that detune the qubit by $\Delta f_Q = 800$ MHz Fig. S2. The contribution of the individual turn-off transients $\varepsilon_{\tau, \text{Off}}$ and $\varepsilon_{\tau + \Delta\tau, \text{Off}}$ is typically significantly larger than $\overline{\Delta f_Q}$ as it takes multiple $\Delta\tau$ for the qubit to return to the sweetspot. However, their difference ε is smaller; with the exception of the first few samples, $\varepsilon \lesssim 8 \text{ MHz} \approx \overline{\Delta f_Q} \cdot 10^{-2}$ [Fig. S2(b, c)]. We next use the acquired phases as input for the Cryoscope analysis to obtain the reconstructed flux $\Phi_R(t)$ and compare it to the true flux $\Phi_Q(t)$ [Fig. S2(d, e)]. We observe a matching of $\Phi_R(t)$ to $\Phi_Q(t)$ better than 1% for $t > 3$ ns. Note that the data shown in Fig. S2(d, e) is normalized to the maximal flux.

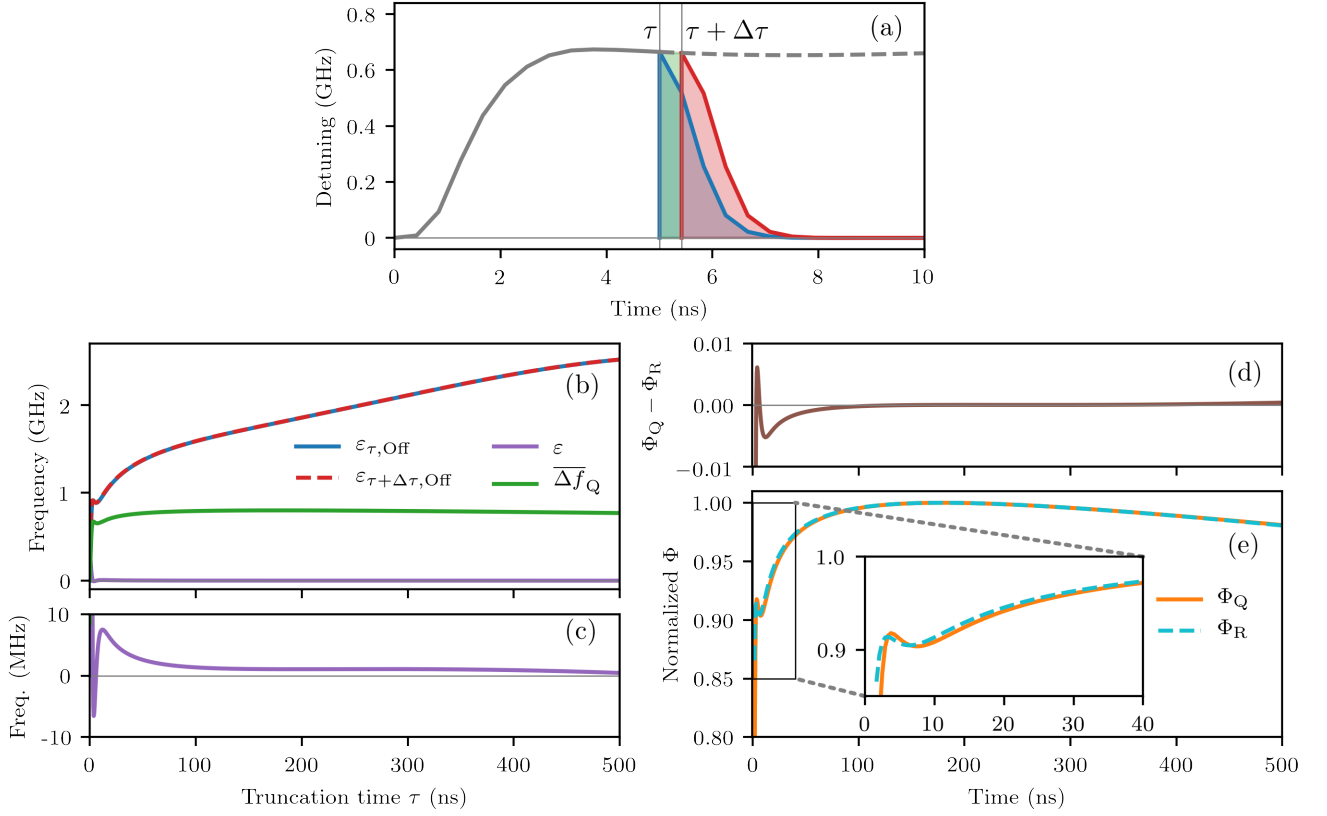


FIG. S2. Simulated reconstruction of a typical step response (cumulative effect of models of Table S1) using Cryoscope. (a) Detuning of the qubit when applying square pulses truncated at τ and $\tau + \Delta\tau$. Shaded areas illustrate the contributions to Δf_R from Δf_Q (green), $\varepsilon_{\tau, \text{Off}}$ (blue) and $\varepsilon_{\tau+\Delta\tau, \text{Off}}$ (red). (b, c) Contributions to Δf_R from the true detuning (green), the individual turn-off transients (blue and red), and the difference of turn-off transients (purple). (e, f) Comparison of the reconstructed flux $\Phi_R(t)$ to the true flux $\Phi_Q(t)$.

B. Cryoscope reconstruction of a single-pole low-pass filter step response

For completeness, we demonstrate that Cryoscope may reconstruct the step response poorly for specific filters. A simple example is the single-pole low-pass filter, whose step response is

$$s_{\text{LP}}(t) = (1 - e^{-t/\tau_{\text{LP}}}) \cdot u(t), \quad (\text{S3})$$

where τ_{LP} is the time constant. Such a filter does not accurately represent our setup but is easy to describe analytically and therefore is a good choice to demonstrate the origin of potential errors and to show the relevance of the nonlinear qubit response to flux. We find that the reconstructed step response $s_{R, \text{LP}}(t)$ differs from $s_{\text{LP}}(t)$ by more than 1% for $t \lesssim 4\tau_{\text{LP}}$.

We also use this simple example to show that Cryoscope is more accurate for higher degrees of nonlinearity. Specifically, we calculate $s_{R, \text{LP}}(t)$ for different forms of qubit detuning on flux: $\Delta f(\Phi) = a\Phi^k$ where $k \in \mathbb{Z}^+$.

In general, the phase φ_τ (setting $T_{\text{sep}} = \infty$) expressed in terms of the impulse response $h = ds/dt$ is

$$\varphi_\tau = 2\pi a \int_0^\infty \left[\int_0^\infty h(t-t') dt' - \int_0^\infty h(t-\tau-t') dt' \right]^k dt = 2\pi a \int_0^\tau \left[\int_0^t h(t-t') dt' \right]^k dt + 2\pi a \int_\tau^\infty \left[\int_0^\tau h(t-t') dt' \right]^k dt, \quad (\text{S4})$$

while the reconstructed step response is given by

$$s_R(\tau) = \left(\frac{2\pi}{a} \frac{d\varphi}{d\tau} \right)^{1/k} = \left(\int_\tau^\infty \left\{ h(t-\tau) \left[\int_0^t h(t-t') dt' \right]^{k-1} \right\} dt \right)^{1/k}. \quad (\text{S5})$$

For the single-pole low-pass filter,

$$s_{R,LP}(t) = \left[1 - e^{-t/\tau_{LP}}\right]^{\frac{k-1}{k}} u(t) = \begin{cases} (1 - e^{-t/\tau_{LP}}) \cdot u(t) & \text{for } k \rightarrow \infty \\ u(t) & \text{for } k = 1 \end{cases} \quad (S6)$$

We observe that, $s_{R,LP}$ matches s_{LP} in the limit $k \rightarrow \infty$. On the other hand, for $k = 1$ (i.e. linear dependence of qubit frequency on flux) the reconstruction gives $u(t)$, completely missing the dynamics.

By directly comparing Eq. (S3) to Eq. (S6) we can place an upper bound on the Cryoscope inaccuracy:

$$0 \leq s_{R,LP}(t) - s_{LP}(t) \leq \frac{1}{k} e^{-t/\tau_{LP}}. \quad (S7)$$

The factor $1/k$ clearly shows that the nonlinear dependence on flux is essential for Cryoscope to work and that a higher power dependence increases its accuracy.

The reduced inaccuracy with higher order k can intuitively be understood by considering that ε is determined by contributions during the turn-off transients (see Eq. (S2)). In the case of a linear dependence, i.e., $k = 1$, contributions during the turn-off transients are weighted as strong as contributions during the pulse, making it impossible to reconstruct the waveforms. With a parabolic dependence, i.e., $k = 2$, the contributions during the off-transients are suppressed. The higher the order k , the more strongly ε is suppressed.

III. CRYOSCOPE SIGNAL-TO-NOISE RATIO

Taking the expectation values $\langle X \rangle$ and $\langle Y \rangle$ in Cryoscope, denoted x and y for brevity, the probability density $p(x, y)$ of measuring (x, y) , given that the true values are (x_0, y_0) , is assumed to be normally distributed with standard deviation σ dependent on readout fidelity and number of averages,

$$p(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-x_0)^2 + (y-y_0)^2}{2\sigma^2}\right). \quad (S8)$$

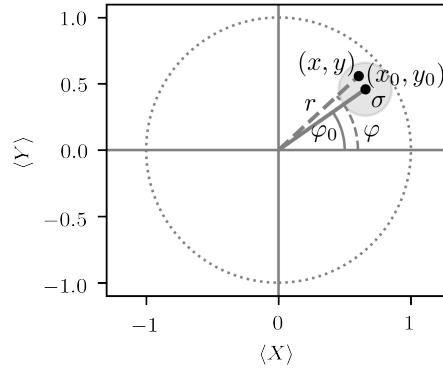


FIG. S3. Illustration clarifying the variables used. The probability of measuring (x, y) , given that the actual values are (x_0, y_0) , is taken to be normally distributed with standard deviation σ denoted by the grey circle.

The values x and y can be represented as an amplitude r and a phase φ . Assuming a perfect (non-distorted) square pulse, the value r is affected by the dephasing of the qubit $r_\tau = \exp(-(\tau/T_2^*(\Phi_Q))^\alpha)$ where $T_2^*(\Phi)$ is the flux dependent dephasing rate and $\alpha \in [1, 2]$. Meanwhile, the phase $\varphi_\tau = 2\pi\tau\Delta f_Q(\Phi_Q)$, where $\Delta f_Q(\Phi_Q)$ is the qubit detuning from the operating point as a function of flux-pulse amplitude. The phase is measured with some error σ_φ that depends on r and σ . Ultimately, to reconstruct Φ_R , we are interested in the phase change between two measurements with slightly different τ , $\Delta\varphi = \varphi_{\tau+\Delta\tau} - \varphi_\tau$, with error $\sigma_{\Delta\varphi}$.

A. Error propagation

Without loss of generality, φ_0 can be set to 0, setting $y_0 = 0$ in Eq. (S8). Rewriting $p(x, y)$ in spherical coordinates,

$$p(r, \varphi) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{r^2 - 2rx_0\cos(\varphi) + x_0^2}{2\sigma^2}\right), \quad (\text{S9})$$

where the subscript from r_τ and φ_τ is dropped for brevity. As we are only interested in the error of the phase, we can integrate to find

$$p(\varphi) = \int_0^\infty p(r, \varphi) r dr = \frac{1}{2\pi} \exp\left(-\frac{x_0^2}{2\sigma^2}\right) + \frac{\cos(\varphi)}{\sqrt{8\pi}} \frac{x_0}{\sigma} \exp\left(-\frac{x_0^2 \sin^2(\varphi)}{2\sigma^2}\right) \left[\operatorname{erf}\left(\frac{x_0 \cos(\varphi)}{\sqrt{2}\sigma}\right) + 1\right]. \quad (\text{S10})$$

When the the visibility of the Ramsey oscillations x_0 is much larger than σ this simplifies to

$$p(\varphi) = \frac{\cos(\varphi)}{\sqrt{8\pi}} \frac{x_0}{\sigma} \exp\left(-\frac{x_0^2 \sin^2(\varphi)}{2\sigma^2}\right) \left[\operatorname{erf}\left(\frac{x_0 \cos(\varphi)}{\sqrt{2}\sigma}\right) + 1\right]. \quad (\text{S11})$$

Because we have set $\varphi_0 = 0$ and $x_0 \ll \sigma$, the small-angle approximation ($\varphi \ll 1$) can be made, so that this simplifies further to

$$p(\varphi) = \frac{x_0}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x_0^2 \varphi^2}{2\sigma^2}\right). \quad (\text{S12})$$

Since the distribution is normal, we conclude that

$$\sigma_\varphi = \frac{\sigma}{x_0} \quad \text{and} \quad \sigma_{\Delta\varphi} = \frac{\sqrt{2}\sigma}{x_0}. \quad (\text{S13})$$

B. SNR formula

Ultimately, the SNR of Cryoscope is affected by the following factors:

1. Readout fidelity and averaging, captured by σ ;
2. Flux dependent qubit dephasing $T_2^*(\Phi_Q)$, affecting visibility $r = x_0 = \exp(-(\tau/T_2^*(\Phi_Q))^\alpha)$, with $1 \leq \alpha \leq 2$;
3. Rate at which the phase is acquired, proportional to $\Delta f_Q(\Phi_Q)$;
4. Filtering effects in the data processing, that affect SNR linearly.

The phase $\Delta\varphi$ in the time interval $\Delta\tau$ is

$$\Delta\varphi = 2\pi\Delta f_Q(\Phi_Q)\Delta\tau, \quad (\text{S14})$$

while the noise of the phase measurement is

$$\sigma_{\Delta\varphi} = \frac{\sqrt{2}\sigma}{x_0}, \quad (\text{S15})$$

leading to

$$\text{SNR} = c' \times \frac{2\pi \times \Delta f_Q(\Phi_Q) \times \Delta\tau \exp(-(t/T_2^*(\Phi_Q))^\alpha)}{\sqrt{2}\sigma}. \quad (\text{S16})$$

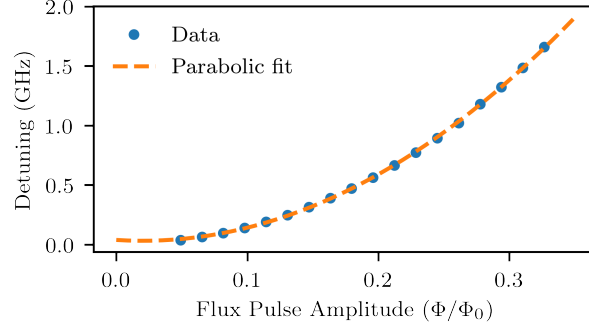


FIG. S4. Measured dependence of detuning on the applied flux through the SQUID loop of the studied transmon qubit.

Here, c' is a constant that accounts for filtering effects in data processing. As we cannot distinguish between σ and effects of filtering c' , and c' is unknown, we can absorb all multiplicative factors

$$\text{SNR} = c'' \times \Delta f_Q(\Phi_Q) \exp(-(\tau/T_2^*(\Phi_Q))^\alpha). \quad (\text{S17})$$

To evaluate the model, we use a quadratic dependence of the detuning on the flux $\Delta f_Q(\Phi_Q) = a\Phi_Q^2$. We find that this dependence matches well the experimentally measured dependence in studied range of pulse amplitudes, up to $0.33 \Phi_0$ for $a = 16.9 \frac{\text{GHz}}{\Phi_0^2}$ (Fig. S4). Furthermore, we use $\alpha = 1$ and a dephasing rate given by

$$\frac{1}{T_2^*(\Phi_Q)} = \Gamma = \Gamma_0 + \Gamma_1 \left| \frac{d\Delta f_Q}{d\Phi_Q} \right|, \quad (\text{S18})$$

where Γ_0 describes the flux-independent dephasing, and Γ_1 parametrizes the contribution to dephasing due to $1/f$ flux noise^{S3}. Finally, in the limit of small errors:

$$\text{SNR} = c\Phi_Q^2 \exp(-(\Gamma_0 + 2a\Gamma_1\Phi_Q)t). \quad (\text{S19})$$

The joint fit to all data in Fig. 3 uses $\Gamma_0 = 66.7 \cdot 10^{-3} \text{ s}^{-1}$, corresponding to the measured sweetspot $T_2^* = 15 \mu\text{s}$ and yields $\Gamma_1 = 0.213 \times 10^{-3} \Phi_0$ equivalent to $\sqrt{A_\Phi} = 12 \times 10^{-6} \Phi_0$ for the single-sided flux noise power spectrum $S(f) = A_\Phi/f$ in a reasonable agreement with the typically reported values^{S4–S7} and consistent with values measured in our group^{S8,S9}.

IV. REAL-TIME PREDISTORTION FILTERS

We make use of two types of digital filters to correct for distortions in real time: a finite impulse response (FIR) filter for short-timescale ($< 30 \text{ ns}$) distortions and a first-order infinite impulse response (IIR) filter.

A. FIR filter

A FIR filter implements a convolution with the impulse response $h_{\text{FIR}}[n] = b_n$ specified by the coefficients b_i and described by:

$$y[n] = \sum_{i=0}^N b_i x[n-i], \quad (\text{S20})$$

where $x[n]$ is the signal at time n at the input of the filter and $y[n]$ is the signal at the output of the filter.

The real-time FIR filter allows specifying 40 parameters to determine the first 72 coefficients b_i . The first 8 parameters directly correspond to the first 8 b_i coefficients while the remaining 32 parameters set pairs of parameters. Allowing a total of 72 coefficients b_i to be set, corresponding to a filter length of 30 ns for the AWG sampling rate of 2.4 GSa/s.

B. Exponential over- and under-shoot correction IIR filter

An IIR filter is slightly more complicated than a FIR filter because it includes feedback:

$$a_0 y[n] = \sum_{i=0}^N b_i x[n-i] - \sum_{i=1}^M a_i y[n-i]. \quad (\text{S21})$$

Here, a_i are the feedback coefficients that describe how $y[n]$ depends on values at the output of the filter at preceding times. Equation (S21) is known as the time-domain difference equation and is a generalization of Eq. (S20).

A first-order IIR filter is implemented in hardware and intended to correct an exponential over- or undershoot in the step response. For a step response described by

$$s(t) = g(1 + Ae^{-t/\tau_{\text{IIR}}}) \cdot u(t), \quad (\text{S22})$$

where τ_{IIR} is the time constant, A the amplitude and g is a gain correction factor that is ignored. The filter that corrects for this effect is described by the coefficients

$$b_0 = 1 - k + k \cdot \alpha, \quad b_1 = -(1 - k) \cdot (1 - \alpha), \quad a_0 = 1 \quad \text{and} \quad a_1 = -(1 - \alpha),$$

with

$$\alpha = 1 - e^{1/f_s \tau_{\text{IIR}}(1+A)} \quad \text{and} \quad k = \begin{cases} \frac{A}{(1+A)(1-\alpha)}, & \text{if } A < 0 \\ \frac{A}{(1+A-\alpha)}, & \text{if } A \geq 0, \end{cases}$$

where f_s is the sampling rate and $a_i, b_i = 0$ for $i > 1$.

The limitations of the hardware implementation of the IIR filter can best be described using an equivalent representation of the filter. The ideal IIR filter obeys the following difference equation mapping the input samples $x[n]$ to the output samples $y[n]$

$$y[n] = (1 - k)x[n] + ku[n], \quad (\text{S23})$$

where $u[n]$ represents the state of the IIR filter, which is determined by the recursive difference equation, known as an exponential moving average

$$u[n] = u[n-1] + \alpha(x[n] - u[n-1]). \quad (\text{S24})$$

Implementing the recursion in Eq. (S24) directly with state-of-the-art digital signal processing hardware is infeasible due to the high sampling rate (2.4 GSa/s). Instead, the real-time filters compute the state variable $u[n]$ based on an average of 16 samples. Furthermore, the IIR filter is operated at a clock frequency of 300 MHz, which means that the state variable $u[n]$ gets updated only every 8-th sample. The down sampled $u[n]$ is combined with the input signal $x[n]$ in Eq. (S23) at the full sampling rate. These hardware approximations were taken into account when modeling the impulse response of the filter.

C. Modelling filters

The Python library SciPy^{S10} provides a function called “lfilter(b, a, sig)”, which applies the filter defined by the coefficient vectors “b” and “a” to the signal defined by the vector “sig”. In this work, we use this function to predict the effect of applying the real-time predistortion filters. Because the implementations of the real-time digital filters in hardware requires certain modifications to approximate the ideal filter operation there are slight differences between the real-time filters and the ideal filters. These deviations are taken into account when we predict the effect of applying a specific filter.

V. USING CRYOSCOPE TO MEASURE ARBITRARY SHAPES

Cryoscope is capable of sampling arbitrary flux control pulses. To demonstrate this capability we have chosen a typical Amsterdam canal skyline as an example of an arbitrary shape for the flux pulse. Figure S5 demonstrates this capability by providing a near perfect reconstruction of the target waveform. Note that the reconstruction involves no free parameters.

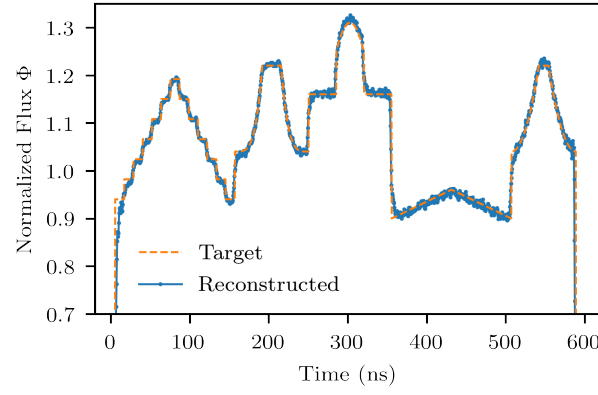


FIG. S5. Reconstruction of an arbitrary waveform consisting of a typical Amsterdam canal skyline.

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